|STORAGE QA ENGINEER | PYTHON AUTOMATION|

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SUMMARY:

Storage QA Engineer with **3+ years** of experience in enterprise storage validation, Python automation, and firmware testing. Skilled in Pytest, Linux, RAID, SAN, Fibre Channel (FC), iSCSI, NVMe-oF, BIOS/BMC/CPLD validation, Jenkins, Git, and performance benchmarking using FIO. Experienced in automation framework development, log analysis, and AI-assisted development using GitHub Copilot.

Professional Experience:

- **3+ years of experience in manual & automation testing on enterprise Storage Arrays**
- OS Experience: Linux, Windows, VMware (vSphere, vCenter, Clusters)
- Python Scripting: OOPs, custom utilities, test frameworks using Pytest
- Designed modular Pytest suites with reusable fixtures to speed up automation and minimize code duplication
- Worked with modules: logging, argparse, paramiko, os, re, time, pickle, json, csv
- Strong experience writing, debugging, and running scripts in PyCharm IDE
- Hands-on RAID (1,5,6) configuration, LUN management, and snapshot testing
- Automated test cases for replication, remote copy, deduplication, and failover
- Performed storage performance tests using FIO (IOPS, Latency, Throughput)
- Protocols: FC, iSCSI, NVMe-oF; Topologies: DAS, SAN, NAS
- Validated NVMe SSDs for signal integrity and hot-swap during platform testing
- Flash/Upgrade firmware using StoreCLI, DUU (vendor tools)
- Validated SVOS-managed features like RAID, DRR, tiering, and snapshots using RAIDCOM CLI on SVP
- Firmware validation: BIOS, BMC, CPLD, PSU, Fans, Backplane
- Used GitHub, Bitbucket, Git Bash for version control
- Leveraged AI**-assisted tools such as GitHub Copilot and Claude** for Python/Pytest automation, test case generation, log analysis, root-cause analysis, troubleshooting, improving development efficiency and reducing test development time.
- Hands-on in Agile workflows: Epic, User Story, Sprint, Scrum, Retrospective
- VMware Features: HA, FT, DRS, Snapshots, Clones, Templates
- Performed **NVMe SSD and PCIe validation** covering drive enumeration, hot-plug/hot-swap, firmware consistency, signal integrity, and PCIe connectivity across storage controllers and backplanes.

Technical Skills:

- **Operating Systems** : Linux(Ubuntu 22.04, RHEL 8/9), Windows Windows Server 2019/2022, VMware ESXi 7.x
- **Programming & Scripting** : Python 3 (OOPs, Automation, Pytest)
- **Test & Performance Tools** : FIO, IOmeter, PyCharm
- **Utilities & Access Tools** : Putty, WinSCP, SSH, vSphere Web/Client
- **Test Management & Defect Tracking** : Jira (Test case management, bug tracking)
- **Protocols & Interfaces** : Fibre Channel (FC), iSCSI, NVMe-oF
- **Version Control** : GitHub, Bitbucket, Git Bash

PROJECT EXPERIENCE:

PROJECT :1

Title : Storage Platform Validation – Hitachi Vantara VSP 790H
Company name : Intenso Tech Solution
Duration : JUNE 2023
Client : Hitachi Vantara

Description:

The Hitachi Virtual Storage E790H is a midrange enterprise storage system designed for high performance and reliability. I performed functional testing of BIOS, BMC, and CPLD, validated key firmware features like remote management, sensor monitoring, event logging, and boot configuration, and executed firmware upgrade testing across multiple hardware setups to ensure stable integration and system performance.

Roles and Responsibilities:

- Configured and set up test environments for **Hitachi VSP 790H storage systems**, including installation and firmware upgrades for critical components such as **BMC, BIOS, drives, and CPLD** to ensure system stability and validation readiness
- Created manual test cases and automated them using **Python and Pytest framework**.
- Validated firmware update process across multiple operating systems including Linux, Windows, and VMware ESXi to ensure compatibility and stability.
- Conducted regression testing to ensure firmware updates did not affect existing system functionalities and maintained system stability.
- Tested BIOS compatibility with different storage devices including **HDD, SSD and NVMe**, ensuring consistent performance and functionality after functionality after BIOS and firmware updates
- Performed NVMe-oF validation to ensure low-latency access and high availability in SAN environments

- Used Git/Bitbucket for version control and Jenkins pipelines to trigger tests and analyze logs

PROJECT :2

Title :	**:**	Feature Validation of Hitachi VSP 1090E- Series Storage Array
Company name	**:**	Intenso Tech Solution
Duration	**:**	MARCH 2025 – Present
Client	**:**	Hitachi Vantara (via Vendor)

Description:

Validated key features of the Hitachi VSP 1090 Series storage array including RAID, snapshots, replication, and failover; automated test cases using Python (Pytest) and performed performance benchmarking with FIO (IOPS, latency, throughput) across FC, iSCSI, and NVMe setups

Roles and Responsibilities:

- Performed feature validation of the Hitachi Vantara VSP 1090 all-flash enterprise storage platform to verify storage reliability, performance, feature functionality
- Developed and executed Python automation test scripts using the Pytest framework for storage feature and platform validation.
- Installed and configured test environments and infrastructure to support efficient storage validation and test execution.
- Configured and validated RAID levels including RAID 1, RAID 5, RAID 6, and RAID 10
- Created RAID arrays, storage pools, and mapped LUNs to hosts for efficient storage management
- Analyzed infrastructure requirements including HA, RAID configurations, LUN Provisioning management
- Validated volume operations including creation, resizing, snapshot management, and deletion
- Designed and executed manual and automated test cases using Python and Pytest
- Performed multipathing validation to verify path failover and load balancing across Fibre(FC), iSCSI, and NVMe-oF environments.
- Improved issue detection rate by 30% through enhanced validation and automation scripts
- Utilized Git and Bitbucket for version control, and leveraged Jenkins pipelines to execute automated tests and log analysis

Educational Qualification:

Bachelor of Technology (B.Tech) in Electronic and Communication Engineering - Bharath University, Chennai| CGPA: 8.7/10